

Derivative Pricing

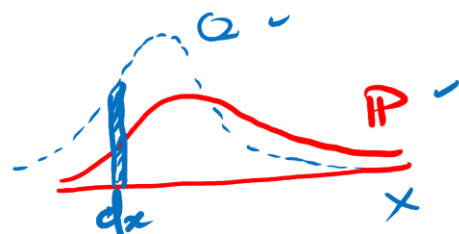
(4. Change of Measure - Continuous)

From Discrete to Continuous

We defined Radon-Nikodym derivative (ζ) defined as the ratio of path probabilities under $\mathbb{Q}$ and under $\mathbb{P}$. In discrete world, it was simply the ratio of point masses. Now we will work in continuous domain.

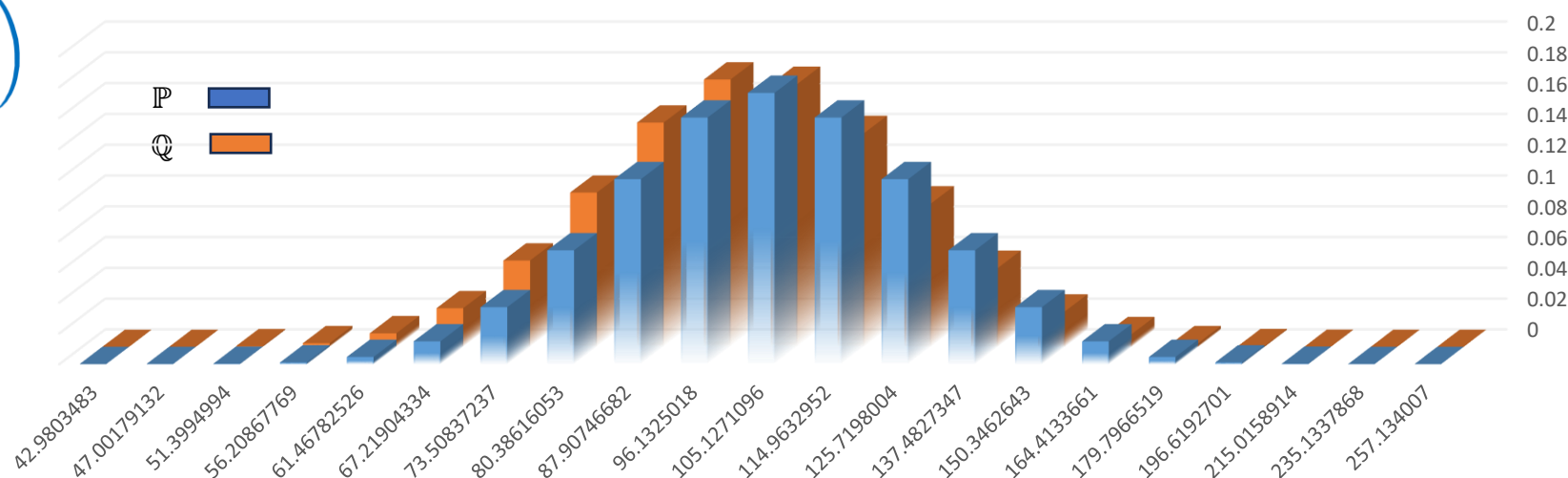
X_t : process
 $\mathbb{P}, \mathbb{Q}$: measure

$$\zeta = \left(\frac{d\mathbb{Q}}{d\mathbb{P}} \right)$$



$$\zeta = \frac{d\mathbb{Q}}{d\mathbb{P}}$$

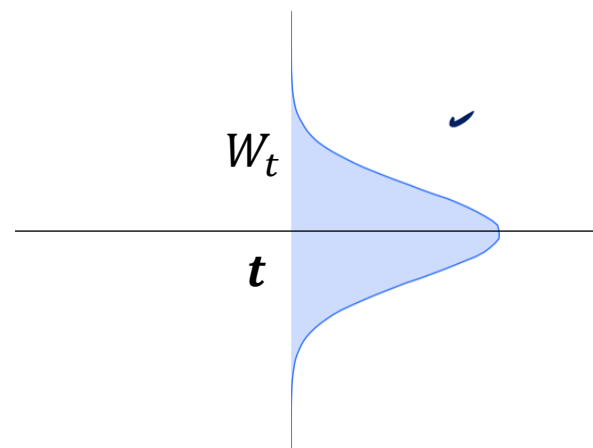
$$\mathbb{E}^{\mathbb{Q}}$$



$\mathbb{P}$ – Brownian

In continuous world, the randomness is captured by Brownian Motion (diffusion models). Consider a $\mathbb{P}$ – Brownian W_t . What this means is W_t has the following distribution under $\mathbb{P}$ measure

$$W_t \sim N(0, t) \quad (\text{under } \mathbb{P})$$



$$f^{\mathbb{P}}(x) = \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{x^2}{2t}\right) \quad (\text{density under } \mathbb{P})$$

How does the distribution of W_t look like under an equivalent measure $\mathbb{Q}$? →

Moment-Generating Functions (I)

To answer the question, let's take a detour and discuss a very useful tool called **Moment generating functions** (MGF) which is an alternative form of expressing probability distribution. The MGF as the name suggests is used to generate moments of a random variable analytically. The MGF of X is given as

$$\textcircled{X} \quad \underline{M_X(\theta)} = \mathbb{E}[e^{\theta X}]$$

which is the expectation of $e^{\theta X}$, where θ is a deterministic parameter. Using Taylor expansion and the linear property of expectation

$$\sigma^2 = \mathbb{E}((X - \mu)^2)$$

$$m_2 = \mathbb{E}(X^2)$$

ith (m_i)

$$\rightarrow e^{\theta X} = 1 + \theta X + \frac{1}{2!} \theta^2 X^2 + \frac{1}{3!} \theta^3 X^3 + \dots$$

$$\underline{\mathbb{E}[e^{\theta X}] = 1 + \theta \mathbb{E}[X] + \frac{1}{2!} \theta^2 \mathbb{E}[X^2] + \frac{1}{3!} \theta^3 \mathbb{E}[X^3] + \dots}$$

$$\underline{\mathbb{E}[e^{\theta X}] = 1 + \theta m_1 + \frac{1}{2!} \theta^2 m_2 + \frac{1}{3!} \theta^3 m_3 + \dots}$$

$$\textcircled{m_1} + \frac{1}{2!} \theta^2 m_2 + \dots$$

$$\frac{1}{2!} \theta^2 m_2 + \dots$$

$$\textcircled{m_2}$$

Differentiating the expression above i times w.r.t to θ and putting $\theta = 0$ gives the i th moment of X around 0. What is the MGF of a normal distribution? ✓



Moment-Generating Functions (II)

Consider a normal random $X \sim N(\mu, \sigma^2)$ under $\mathbb{P}$ measure. It's moment generating function is given as

$$M_X(\theta) = \mathbb{E}[e^{i\theta X}] = \exp\left(\theta \mu + \frac{1}{2} \theta^2 \sigma^2\right)$$

Handwritten notes: A blue arrow points from the 'i' in the exponent to the 'theta' in the argument. The terms 'theta mu' and '1/2 theta^2 sigma^2' are underlined. A red arrow points to the right of the exponential function. There are also some scribbles to the left of the equation.

The moment generating function pins down the probability distribution. This means if the MGFs of two random variables are same, then the probability distributions will also be same.

There is a related concept called characteristic function. We will revisit this later during Fourier Transform methods of option pricing.

Prove that sum of two normal randoms is a normal random for the independent case.

List of moment generating functions in Wikipedia

https://en.wikipedia.org/wiki/Moment-generating_function

Q – Brownian(I)

Let's come back to the original question. How does a $\mathbb{P}$ – Brownian W_T look like under measure $\mathbb{Q}$? For that we need to define $\mathbb{Q}$. Let's define the linkage between $\mathbb{P}$ and $\mathbb{Q}$ via a suitable Radon-Nikodym derivative. Let's define the T -horizon RN derivative as the following function



$$\frac{d\mathbb{Q}}{d\mathbb{P}} = \exp\left(-\lambda W_T - \frac{1}{2}\lambda^2 T\right)$$

The above choice of RN derivative will be clear very soon. To find the distribution W_T under $\mathbb{Q}$, we can take the MGF route. We know the expectation of a random variable under $\mathbb{Q}$ is same as the expectation under $\mathbb{P}$ of the random variable deflated by the RN derivative.

$$\int e^{\theta W_T} d\mathbb{Q} = \int e^{\theta W_T} \frac{d\mathbb{Q}}{d\mathbb{P}} d\mathbb{P}$$

$$\mathbb{E}^{\mathbb{Q}}[e^{\theta W_T}] = \mathbb{E}^{\mathbb{P}}\left[e^{\theta W_T} \frac{d\mathbb{Q}}{d\mathbb{P}}\right]$$

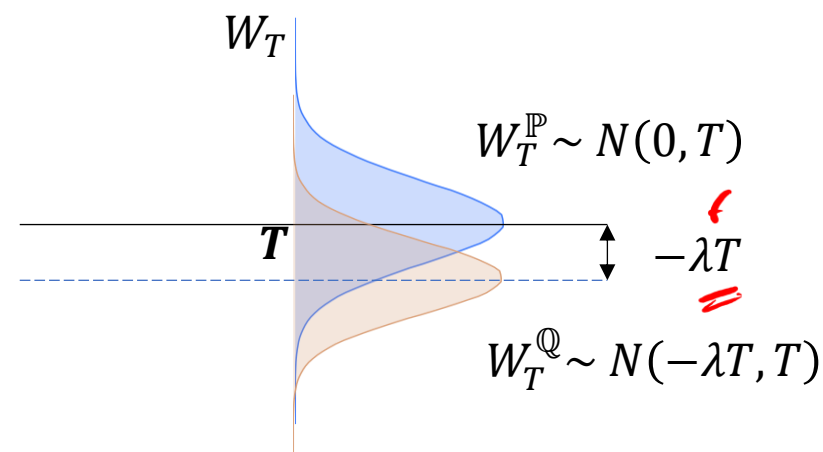
Prove that this expression simplifies to $\mathbb{E}^{\mathbb{Q}}[e^{\theta W_T}] = \exp\left(-\theta \lambda T + \frac{1}{2}\theta^2 T\right)$

$\mathbb{Q}$ – Brownian(II)

Comparing the MGF with the formula for Normal distribution, it's clear that

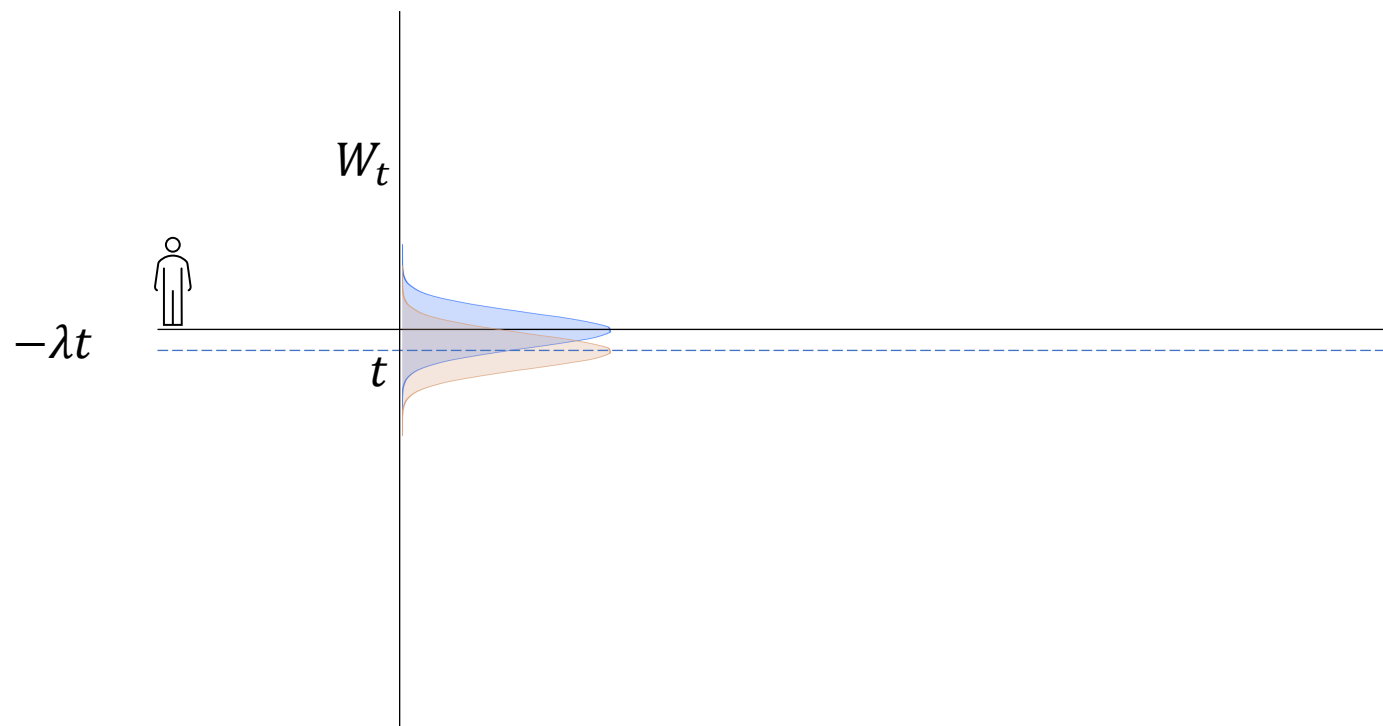
$$\underline{\underline{W_T}} \sim N(\underline{\underline{-\lambda T}}, T) \quad (\text{under } \underline{\underline{\mathbb{Q}}})$$

Note that under $\mathbb{Q}$, $\underline{\underline{W_T}}$ is still normal. The variance is unaffected, only the mean is distorted. The RN derivative does something special. It drags the distribution down by an amount λT after time T or in more general terms by an amount λt at time t

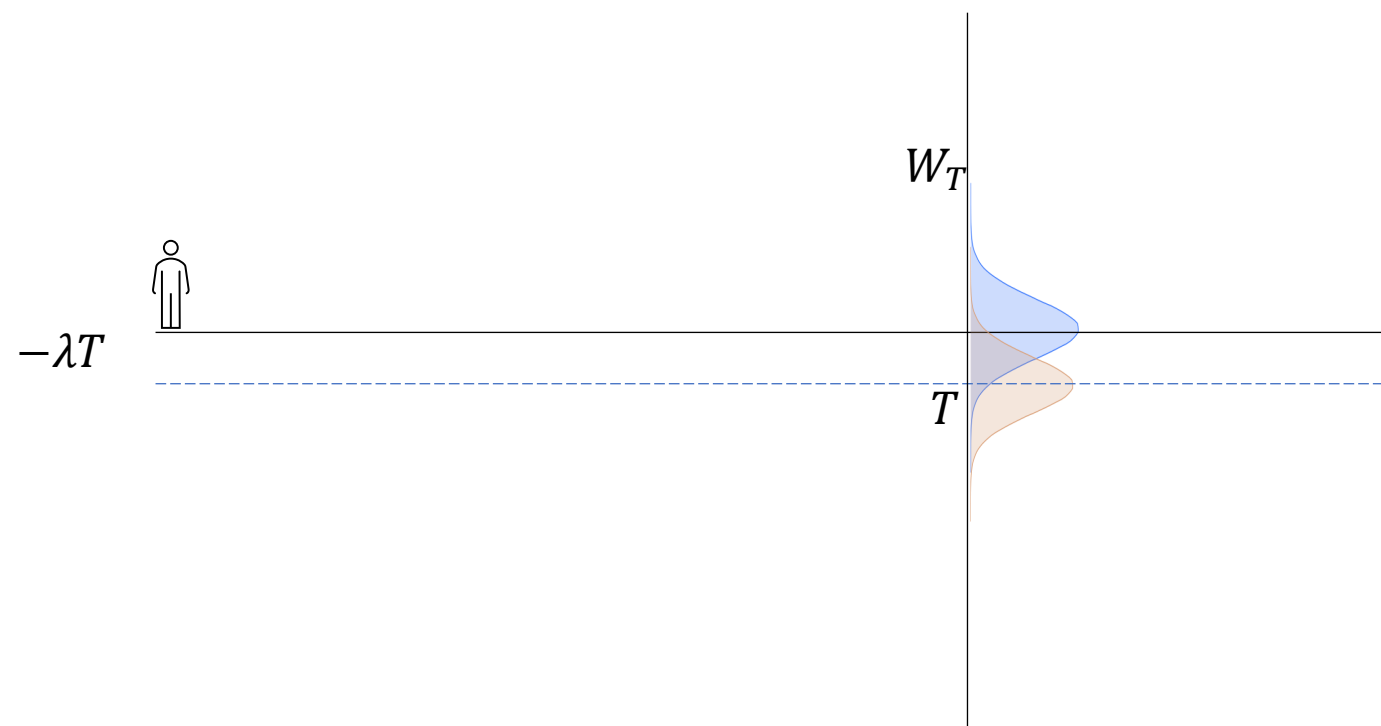


Show that $\frac{d\mathbb{Q}}{d\mathbb{P}} = \exp\left(-\lambda W_T - \frac{1}{2}\lambda^2 T\right)$ using pdf of normal distribution.

$\mathbb{P}$ – Brownian vs $\mathbb{Q}$ – Brownian



$\mathbb{P}$ – Brownian vs $\mathbb{Q}$ – Brownian



Q- Brownian (III)

To summarize,

- We started with W_t which is a **Standard Brownian** (mean = 0) under measure $\mathbb{P}$
- We found a specific $\mathbb{Q}$ measure, under which W_t has a mean $\neq 0$ but variance remains intact
- The measure change is captured by RN which has a specific form

$$W_T^{\mathbb{Q}} \sim N(-\lambda T, T)$$

$$t=1 \\ W_1 \sim N(0, 1)$$

Is there another process which is **Standard Brownian** $\mathbb{Q}$ then? $(-\lambda T, T)$

Yes. It's

$$\tilde{W}_t = W_t + \lambda t$$

$$\tilde{W}_t^{\mathbb{Q}} \sim N(0, t)$$

$$W_t^{\mathbb{P}} \sim N(0, t)$$

$$W_T^{\mathbb{Q}} \sim N(-\lambda T, T)$$

$$\tilde{W}_t^{\mathbb{Q}} \sim N(0, t)$$

Or in SDE form

$$d\tilde{W}_t = dW_t + \lambda dt$$

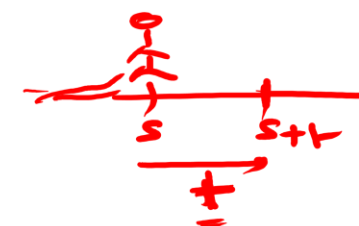
Q- Brownian (II)

Verify the following expressions.

$$\mathbb{E}^{\mathbb{Q}}(\exp(\theta \tilde{W}_t)) = \exp\left(\frac{1}{2}\theta^2 t\right) = e^{(\theta^2 t + \frac{1}{2}\theta^2 t)} = e^{\left(\frac{1}{2}\theta^2 t\right)}$$

$$\mathbb{E}^{\mathbb{Q}}(\exp(\theta(\tilde{W}_{s+t} - \tilde{W}_s)) \mid \mathcal{F}_s) = \exp\left(\frac{1}{2}\theta^2 t\right)$$

$N(0, t)$



Cameron-Martin-Girsanov (CMG)

Cameron-Martin-Girsanov (CMG) theorem

If W_t is a $\mathbb{P}$ -Brownian and λ_t is a pre-visible process which satisfies the bounded condition $\mathbb{E}^{\mathbb{P}} \left(\exp \left(\frac{1}{2} \int_0^T \lambda_t^2 dt \right) \right) < \infty$ (Novikov's condition), then there exists a measure $\mathbb{Q}$ such that

- 1 $\mathbb{Q}$ is equivalent to $\mathbb{P}$ ✓
- 2 $\frac{d\mathbb{Q}}{d\mathbb{P}} = \exp \left(- \int_0^T \lambda_t dW_t - \frac{1}{2} \int_0^T \lambda_t^2 dt \right)$ $\rightarrow$
- 3 $\tilde{W}_t = W_t + \int_0^t \lambda_s ds$ which is a $\mathbb{Q}$ -Brownian

$$0 \rightarrow t \quad e^{-\lambda W_t - \frac{1}{2} \lambda^2 t}$$

CMG – Applications

CMG is a powerful tool to control the drift of a process. Consider the following example

$$dX_t = \mu dt + \sigma dW_t, \quad X_t = \mu t + \sigma W_t \quad x_t \sim N(\mu t, \sigma^2 t)$$

where W_t is a $\mathbb{P}$ – Brownian. Under what measure X_t has a drift of r ?

Rearranging the above equation

$$dX_t = \sigma \left(dW_t + \left(\frac{\mu - r}{\sigma} \right) dt \right) + r dt$$

$$d\tilde{W}_t = dW_t + \left(\frac{\mu - r}{\sigma} \right) dt$$

$$dx_t = r dt + \sigma d\tilde{W}_t$$

$\lambda = \frac{\mu - r}{\sigma}$

and $d\tilde{W}_t$ is a $\mathbb{Q}$ – Brownian, where $\mathbb{Q}$ and $\mathbb{P}$ are linked through the RN $\frac{d\mathbb{Q}}{d\mathbb{P}} = \exp\left(-\lambda W_T - \frac{1}{2}\lambda^2 T\right)$ where $\lambda = \left(\frac{\mu - r}{\sigma}\right)$. So, under $\mathbb{Q}$, X_t has a different drift of r , $dX_t = r dt + \sigma d\tilde{W}_t$

Recognize the structure of λ ?

Exercises

1. Let $X_t = \sigma W_t + \mu t$, where W_t is a $\mathbb{P}$ -Brownian. Find the measure $\mathbb{Q}$ under which X_t is a martingale.
2. Let $dS_t = \sigma S_t dW_t + \mu S_t dt$, where W_t is a $\mathbb{P}$ -Brownian. Find the measure $\mathbb{Q}$ under which X_t has a drift r .
3. If W_t is a $\mathbb{P}$ -Brownian, and $\frac{d\mathbb{Q}}{d\mathbb{P}} = \exp\left(W_t - \frac{1}{2}t\right)$, then find the $\mathbb{Q}$ -Brownian $\tilde{W}_t$.
4. Show that $\zeta_t = \exp\left(-\lambda W_t - \frac{1}{2}\lambda^2 t\right)$ is a martingale under $\mathbb{P}$. What is the SDE of ζ_t ?

Home-Work

1. Let $X \sim N(0,1)$ ^{under $\mathbb{P}$} under $\mathbb{P}$, and $(a, \alpha, \beta) \in \mathbb{R}$. Define a new measure $\tilde{\mathbb{P}}$ by

$$d\tilde{\mathbb{P}} = \exp(\alpha X + \beta) d\mathbb{P}$$

Find (α, β) in terms of a such that $X + a \sim N(0,1)$ under $\tilde{\mathbb{P}}$

2. Suppose M is a continuous process such that both M and M^2 are martingales. Must M be constant in time? Prove it, or find a counter example

Martingale Representation Theorem

Suppose M_t and N_t are both $\mathbb{Q}$ – martingale processes, then there exists a pre-visible process ϕ such that N_t can be written as

$$\underline{N_t} = \underline{N_0} + \int_0^t \underline{\phi_s} d\underline{M_s}$$

$$N_t = N_0 + \sum_{i=1}^t \phi_i \Delta M_i$$

ϕ_s is unique which is the portfolio process. Notice that we discussed this in discrete world where we had the sum which have been replaced by integral in the continuous world.